

Tanjim Ahmed

EEE Undergraduate | RTL Design & VLSI Enthusiast | Semiconductor Researcher
American International University-Bangladesh
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About Me

I am Tanjim Ahmed, an 11th-semester Electrical and Electronic Engineering student at American International University-Bangladesh. I have a strong interest in the semiconductor field, especially RTL design, VLSI, SystemVerilog, and device-level simulation.

I am currently working on digital design projects using RTL and SystemVerilog, and I am also involved in research related to photovoltaic, semiconductor device modeling and Communication. My research work focuses on SCAPS-1D-based solar cell simulation, lead-free perovskite materials, and performance optimization of high-efficiency photovoltaic devices.

Along with semiconductor and VLSI interests, I have programming knowledge in Python, C, and C++. I am also exploring simulation tools such as Silvaco TCAD and SCAPS-1D for device modeling and analysis.

Technical Skills

Semiconductor & VLSI:

RTL Design, VLSI Design, SystemVerilog, Digital Logic Design, Design Verification basics, IC Physical Design basics, Analog Design basics

Programming:

Python, C, C++

Simulation & Research Tools:

SCAPS-1D, Silvaco TCAD, MatLab

Research Area:

Perovskite Solar Cells, Lead-free Solar Cell Modeling, Semiconductor Device Simulation, Photovoltaic Performance Optimization, semiconductor materials, communication (Semantic Communication).

Research Experience

Researcher - Perovskite Solar Cell Simulation

Currently involved in a journal research work on an eco-friendly high-efficiency double absorber perovskite solar cell using Cs_2TiBr_6 and $\text{La}_2\text{NiMnO}_6$ through SCAPS-1D modeling. The paper investigates lead-free double absorber solar cell architecture and reports optimized photovoltaic parameters including PCE, Voc, Jsc, and FF. My contribution includes conceptualization, data

curation, formal analysis, software-based simulation support, resources, and review/editing of the manuscript.

Status: Journal manuscript under 3rd review.

Training & Certifications

Industrial Training - ULKASEMI Pvt. Limited

Completed a 9-day industrial training program conducted by ULKASEMI Pvt. Limited. The training covered Design Verification, Analog Design, IC Physical Design, Foundry Technologies, IC Mask Design, CV Writing & Interview Skills, Presentation Skills, and Public Speaking.

Workshop on VLSI Design and Semiconductor Industry - ChipXpert Technologies Private Limited

Participated in a two-day workshop on VLSI Design and the Semiconductor Industry organized by ChipXpert Technologies Private Limited.

Projects

Vending Machine Design - Ongoing

Currently working on an RTL-based vending machine design project. The project focuses on digital logic design, finite state machine implementation, and SystemVerilog-based RTL development.

RTL & SystemVerilog Practice Projects

Working on digital design concepts including FSM design, datapath design, testbench development, and simulation-based verification.

Portfolio Short Bio

I am an EEE undergraduate student with a strong passion for semiconductor technology, RTL design, VLSI, and device simulation. My interests connect both digital IC design and semiconductor research. I have hands-on experience with SystemVerilog, RTL design, SCAPS-1D, Silvaco TCAD, Python, C, and C++. I am also involved in journal-level research on lead-free perovskite solar cells and currently working on RTL-based digital design projects.